A proof of the conjectured closed form for OEIS A208545

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Abstract

OEIS A208545 carries a conjectured closed form. We prove it. The entry states, as fact, a generating function, and from that a linear recurrence with polynomial coefficients satisfied by the sequence is derived exactly. The conjectured closed form is then shown to satisfy the same recurrence, and to agree with the sequence at enough consecutive indices. A recurrence of order 5 whose leading coefficient does not vanish, together with 5 consecutive values, determines a sequence completely, so the two agree everywhere. Nothing is checked numerically and then assumed: the recurrence is derived symbolically, the verification is an exact identity, and the finitely many indices where the leading coefficient vanishes are computed rather than waved at.

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1 The sequence, what is known, and what is conjectured

OEIS A208545 is "Number of 7-bead necklaces of n colors allowing reversal, with no adjacent beads having the same color". It has offset 1 and begins

$$a(1), \dots, a(8) \;=\; 0, 0, 9, 156, 1170, 5580, 19995, 58824, \;\dots$$

The entry states, not as a conjecture,

3*x^3*(3 + 28*x + 58*x^2 + 28*x^3 + 3*x^4) / (1 - x)^8

and separately carries the comment

a(n) = (1/14)*n^7 - (1/2)*n^6 + (3/2)*n^5 - (5/2)*n^4 + (5/2)*n^3 - (3/2)*n^2 + (3/7)*n.

As of the "Last modified" line on the live entry (2017-11-11, revision 10) the second statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

2 A recurrence from what is known

The generating function the entry posts is algebraic over $\mathbb{Q}(x)$, hence D-finite: it satisfies a linear differential equation with polynomial coefficients, obtained by differentiating its defining equation inside the field it generates and taking a linear dependence among the derivatives. Reading off the coefficient of each power of x turns that differential equation into a recurrence for the coefficients, which is

$$(9n+9)\ a(n)+(75n+306)\ a(n{+}1)+(90n+1206)\ a(n{+}2)+(936-90n)\ a(n{+}3)+(81-75n)\ a(n{+}4)+(-9n-1 \tag{1}$$

The leading coefficient of (1) is $-9\,(n+2)$. Its only integer zeros are $n \in \{-2\}$, all below the range claimed here.

3 The conjectured description satisfies it

The conjectured closed form is a sum of hypergeometric terms. Substituting it into (1) and grouping the summands into similarity classes – two terms being similar when their quotient is a rational function of n – every class contributes a rational function of n, and each of those cancels to zero identically. Since pairwise dissimilar hypergeometric terms are linearly independent over $\mathbb{Q}(n)$, that is exactly the statement that the closed form satisfies (1).

4 Uniqueness

Lemma 1. *Let $\sum_{j=0}^{r} q_j(n)\, b(n+j) = 0$ be a linear recurrence whose leading coefficient q_r has no zero at any integer $n \geq n_0$. If two sequences both satisfy it for all $n \geq n_0$ and agree at the r consecutive indices $n_0, \ldots, n_0 + r - 1$, they agree at every $n \geq n_0$.*

Proof. Induction. Suppose the two agree at $n_0, \ldots, m + r - 1$ for some $m \geq n_0$. The recurrence at $n = m$ determines $b(m + r)$ from the preceding r values, because $q_r(m)$ is nonzero and may be divided by; both sequences therefore take the same value at $m + r$. □

Theorem 2. *For all $n \geq 6$,*

$$a(n) \;=\; \frac{n^7}{14} - \frac{n^6}{2} + \frac{3n^5}{2} - \frac{5n^4}{2} + \frac{5n^3}{2} - \frac{3n^2}{2} + \frac{3n}{7}$$

Proof. Both sides satisfy (1): the sequence because the recurrence was derived from the entry's own a generating function, and the conjectured closed form by the computation of the previous section. They agree at the 5 consecutive indices beginning $n = 6$, which was checked against the entry's DATA in exact integer arithmetic. The leading coefficient has no zero at any integer $n \geq 6$, so Lemma 1 applies. □

5 Verification

Three checks, independent of one another.

First, the description the entry states as fact was itself confirmed against the entry's DATA before any recurrence was derived from it; a statement that did not reproduce the entry's own terms would not have been used as the basis of anything.

Second, the derived recurrence (1) was evaluated on the published terms in exact integer arithmetic and vanishes at every index where it applies.

Third, the conjectured closed form was evaluated at the first 13 indices and compared term by term with the DATA; the values agree exactly. This is not part of the proof, which is symbolic, but it would catch a transcription error in either statement.

References

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[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011.